

9. (a) In 1915 Alfred Wegener suggested that the Earth's continents were once joined together as one large land mass.



- (i) State **three** pieces of evidence that Alfred Wegener used to support his theory. [3]

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- (ii) Explain why Wegener's theory was not originally accepted by other scientists, but it is today. [2]

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- (b) What type of destructive event is likely to happen at a **conservative** plate boundary? [1]

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Examiner  
only

6

**END OF PAPER**



### FORMULAE FOR SOME COMMON IONS

| POSITIVE IONS |                  | NEGATIVE IONS |                    |
|---------------|------------------|---------------|--------------------|
| Name          | Formula          | Name          | Formula            |
| aluminium     | $\text{Al}^{3+}$ | bromide       | $\text{Br}^-$      |
| ammonium      | $\text{NH}_4^+$  | carbonate     | $\text{CO}_3^{2-}$ |
| barium        | $\text{Ba}^{2+}$ | chloride      | $\text{Cl}^-$      |
| calcium       | $\text{Ca}^{2+}$ | fluoride      | $\text{F}^-$       |
| copper(II)    | $\text{Cu}^{2+}$ | hydroxide     | $\text{OH}^-$      |
| hydrogen      | $\text{H}^+$     | iodide        | $\text{I}^-$       |
| iron(II)      | $\text{Fe}^{2+}$ | nitrate       | $\text{NO}_3^-$    |
| iron(III)     | $\text{Fe}^{3+}$ | oxide         | $\text{O}^{2-}$    |
| lithium       | $\text{Li}^+$    | sulfate       | $\text{SO}_4^{2-}$ |
| magnesium     | $\text{Mg}^{2+}$ |               |                    |
| nickel        | $\text{Ni}^{2+}$ |               |                    |
| potassium     | $\text{K}^+$     |               |                    |
| silver        | $\text{Ag}^+$    |               |                    |
| sodium        | $\text{Na}^+$    |               |                    |
| zinc          | $\text{Zn}^{2+}$ |               |                    |



## Group

3

4



9

2

0

24

|                                                                                                     |    |           |           |           |          |            |            |           |         |           |         |            |          |           |          |           |          |          |        |          |
|-----------------------------------------------------------------------------------------------------|----|-----------|-----------|-----------|----------|------------|------------|-----------|---------|-----------|---------|------------|----------|-----------|----------|-----------|----------|----------|--------|----------|
| <div><div><div><div><div>1</div><div>H</div><div>Hydrogen</div><div>1</div></div></div></div></div> |    |           |           |           |          |            |            |           |         |           |         |            |          |           |          |           |          |          |        |          |
| 7                                                                                                   | Li | 9         |           |           |          |            |            |           |         | 11        | 12      | 14         | 16       | 19        | 20       |           |          |          |        |          |
| Lithium                                                                                             | Be | Beryllium |           |           |          |            |            |           |         | Boron     | Carbon  | Nitrogen   | Oxygen   | Fluorine  | Neon     |           |          |          |        |          |
| 3                                                                                                   | 4  |           |           |           |          |            |            |           |         | 5         | 6       | 7          | 8        | 9         | 10       |           |          |          |        |          |
| 23                                                                                                  | Na | 24        |           |           |          |            |            |           |         | 27        | 28      | 31         | 32       | 35.5      | 40       |           |          |          |        |          |
| Sodium                                                                                              | Mg | Magnesium |           |           |          |            |            |           |         | Aluminium | Silicon | Phosphorus | Sulfur   | Chlorine  | Argon    |           |          |          |        |          |
| 11                                                                                                  | 12 |           |           |           |          |            |            |           |         | 13        | 14      | 15         | 16       | 17        | 18       |           |          |          |        |          |
| 39                                                                                                  | K  | 40        | 45        | 48        | 51       | 52         | 55         | 56        | 59      | 59        | 63.5    | 65         | 70       | 73        | 75       | 79        | 80       | 84       |        |          |
| Potassium                                                                                           | Ca | Calcium   | Scandium  | Titanium  | Vanadium | Chromium   | Manganese  | Iron      | Cobalt  | Nickel    | Copper  | Zinc       | Gallium  | Germanium | Arsenic  | Selenium  | Bromine  | Krypton  |        |          |
| 19                                                                                                  | 20 |           | 21        | 22        | 23       | 24         | 25         | 26        | 27      | 28        | 29      | 30         | 31       | 32        | 33       | 34        | 35       | 36       |        |          |
| 86                                                                                                  | Rb | 88        | 89        | 91        | 93       | 96         | 99         | 101       | 103     | 106       | 108     | 112        | 115      | 119       | 122      | 128       | 127      | 131      |        |          |
| Rubidium                                                                                            | Sr | Strontium | Yttrium   | Zirconium | Niobium  | Molybdenum | Technetium | Ruthenium | Rhodium | Palladium | Silver  | Cadmium    | Indium   | Tin       | Antimony | Tellurium | Iodine   | Xenon    |        |          |
| 37                                                                                                  | 38 |           | 39        | 40        | 41       | 42         | 43         | 44        | 45      | 46        | 47      | 48         | 49       | 50        | 51       | 52        | 53       | 54       |        |          |
| 133                                                                                                 | Cs | 137       | 139       | 179       | 181      | 184        | 186        | 190       | 192     | 195       | 197     | 201        | 204      | 207       | 209      | 210       | 210      | 222      |        |          |
| Caesium                                                                                             | Ba | Barium    | Lanthanum | Hafnium   | Tantalum | Tungsten   | Rhenium    | Osmium    | Iridium | Platinum  | Gold    | Mercury    | Thallium | Lead      | Bismuth  | Polonium  | Astatine | Radon    |        |          |
| 55                                                                                                  | 56 |           | 57        | 72        | 73       | 74         | 75         | 76        | 77      | 78        | 79      | 80         | 81       | 82        | 83       | 84        | 85       | 86       |        |          |
| 223                                                                                                 | Fr | 226       | 227       |           |          |            |            |           |         |           |         |            |          |           |          |           |          | 227      | 226    | 223      |
| Francium                                                                                            | Ra | Radium    | Actinium  |           |          |            |            |           |         |           |         |            |          |           |          |           |          | Actinium | Radium | Francium |
| 87                                                                                                  | 88 |           | 89        |           |          |            |            |           |         |           |         |            |          |           |          |           |          | 89       | 88     | 87       |

## Key

relative atomic mass

| Name | Symbol | $A_r$ |
|------|--------|-------|
|------|--------|-------|

atomic number